# BFOR516 - Advanced Data Analytics for Cybersecurity

Srishti Gupta, Ph.D.

# Introduction

- Dr. Srishti Gupta
- Current Role:
  - Visiting Assistant Professor
  - Massry School of Business, University at Albany
- Educational Background:
  - Ph.D. in Human-Computer Interaction (HCI)
  - Pennsylvania State University
- Research Interests:
  - HCI, Human-Centered AI, Social Computing

# Course Overview

- Machine learning and AI techniques
- Assessment:
  - In-class Labs: 20%
  - Individual Critical Analysis Paper : 30%
  - Group Project: 25%
  - Individual Learning Portfolio: 15%
  - Attendance: 10%

# Course Evaluation and Policies

1. **In-Class Lab (20%):**
   a. Students will engage in small lab activities during each class session, directly based on that day's lecture. These activities are designed to provide hands-on experience and reinforce the concepts discussed in class.
2. **Individual Critical Analysis Paper (30%):**
   a. This is an individual assignment which will be due in the middle of the semester (around Week 10-11). Students will analyze and write a critical review of a research paper that aligns with one of the course's key topics.
   b. Students will select one research paper from an instructor-provided list. Each student will submit an individual critical analysis of the selected paper based on instructor guidelines. Although multiple students may analyze the same paper, all submissions must reflect independent reasoning and original analysis.

# Course Evaluation and Policies

3. **Group Project (25%):**
   - Students will form their own groups by 02/15/2026:
     - https://docs.google.com/spreadsheets/d/1VxclnEBLPkUTEjchTCOqeOduVnLvoN5TL2k0i4uI99g/edit?usp=sharing
     - Find an available group number.
     - Enter your name in the first empty "Member" slot under the selected group. Groups will have 4 members.
     - Once all slots are filled, group formation will close.
   - Student groups will collaboratively analyze 2-3 machine learning models applied to a common dataset or problem scenario.
   - The project emphasizes performance comparison, interpretability, and ethical considerations. Groups will submit a project report and deliver a final presentation summarizing their findings.
   - Each group must justify their model selection, evaluation approach, and interpretation.

# Course Evaluation and Policies

3. **Group Project (25%)(Continued):**
   - The project will be structured around a series of milestones and deliverables, which may include (but are not limited to): project proposal, dataset selection and documentation, intermediate analyses, final written report, and final presentation.
   - Specific deadlines and detailed instructions for each deliverable will be provided during the semester.
   - Last two class meetings will be devoted to formal group presentations of completed projects.

# Course Evaluation and Policies

3. **Individual Learning Portfolio (15%):**
   - Each student will submit a short reflective portfolio documenting their learning throughout the course. The portfolio should include:
     - Key learning moments and challenges encountered
     - How generative AI tools were used, evaluated, and integrated into your work
     - Reflections on what you would do differently in future analyses or projects
4. **Attendance (10%):** Attendance will be taken each class. Students must attend the entire class to receive credit. Partial attendance will not earn points. This ensures engagement and participation.

# Class Policies

## Late Submission (Group Case Study Deliverables):

- **Advance Communication**:
  - If you anticipate submitting late, communicate in advance.
- **Penalty for Late Submission**:
  - For each day late, 10% of the grade is deducted.
  - Example:
    - 1 day late: 10% deduction
    - 2 days late: 20% deduction, etc.

# Attendance and Absence Policies

- Attendance is mandatory, and students are expected to participate in class fully, just as they would in any traditional in-person course. Students must attend the entire class to receive attendance credit.
- Inform the instructor **at least 24 hours in advance** if you plan to miss a class. If an emergency occurs within 24 hours, notify the instructor as soon as possible. Appropriate **documentation is required for emergency absences**. Failure to provide timely notice and documentation may result in an unexcused absence.
- Approved excused absences do not affect attendance credit. Students must make up all missed work within the timeframe set by the instructor.
- Missing class without a valid excuse counts as 0 points for attendance for that session. Repeated unexcused absences (e.g., more than 4) may impact your ability to pass the course.

# LLMs (ChatGPT, LLMs, etc.)

- You may use LLMs for learning, exploration, and as references **only in ways explicitly permitted and instructed by the instructor**. You are not allowed to use LLMs to generate solutions for lab work, assignments, or projects. Doing so will be considered a violation of academic integrity.
- Since there is no formal way to cite LLM generated content, if you use LLMs in your work, always write a **detailed description** of how it was used to support your work, clearly describing which parts were AI generated.

# Using Code from Online Sources

- When writing code, it is common to use parts of code from others' projects.
- You should cite the resources code snippets you use.
- This can be as simple as a comment with the URL of the site where the code is from.
- Not citing sources is a violation of academic integrity and usually a violation of software licenses.

# Citation & Academic Integrity Policy

1. **Independent Writing**
   a. All written work must be your own.
   b. You are expected to articulate your analysis and arguments yourself, even if English is not your strongest language.
   c. You will not be graded down for grammar, syntax, or imperfect English, provided your ideas are original and clearly conveyed.
2. **Required Use of Citations**
   a. You must cite relevant scholarly articles, books, or white papers to support your claims and critical analysis.
   b. Citations demonstrate engagement with existing research and strengthen your argumentation.

# Citation & Academic Integrity Policy (Cntd.)

3. **Use of Language Models (LLMs)**
   a. You may use LLMs (e.g., ChatGPT) for:
      i. Clarifying difficult texts
      ii. Explaining research concepts
      iii. Refining code or technical descriptions
   b. You may NOT use LLMs to:
      i. Draft paragraphs or write your analysis for you
      ii. Generate complete responses or text you submit
      iii. LLMs are a learning aid not a writing substitute.
4. **Citation Style Requirement**
   a. Use APA style for both In-text citations and Reference list / bibliography. Examples and guidelines are available at Purdue OWL: https://owl.purdue.edu/owl/research_and_citation/apa_style/apa_formatting_and_style_guide/index.html

# Office Hours

## Office Hours:

- Wednesday 10:00 am – 11:00 am, Thursday 11:00 am - 12:00 pm in BB318
- Office hours are by default in person. If you want to meet virtually, let me know if advance, and I will set up a Zoom meeting.

## Inquiries and Other Appointments:

- Email: Include course code and section in the subject line
- If no response within 2 business days, please send a follow-up email
- Room 318, Business Building

## Important Notes:

- **Office Hours:**
  - Not for resolving complex, time-consuming issues like debugging code
  - For such issues, use the discussion board or email me in advance so I can review the problem

# Recap: Regression Analysis

# Overview

- Regression analysis is a tool to investigate how two or more variables are related. Quite often we want to see how a specific variable of interest is affected by one or more variables. For example, one may wish to use a person's height, gender, race, etc. to predict a person's weight.
- Example: If you are asked to estimate the weight of a student, what will you use as a point estimate? If I tell you that the height of the student is 70 inches, can you give a better estimate of the person's weight?
  - For the first part, the point estimate would be the average weight (or the median weight) of a student. If you know the student is 70 inches, then, yes, you can give a better estimate of the person's weight, but only if you have some idea about how height and weight are related.

# Overview

- It is important to distinguish between the variable of interest and the variable(s) we will use to predict the variable of interest.
  - **Response Variable** - Denoted, Y, is also called the variable of interest or dependent variable. In the example, weight is the response variable.
  - **Predictor Variable**- Denoted, X, is also called the explanatory variable or independent variable. In the example, height is the predictor.
- When there is only one predictor variable, we refer to the regression model as a simple linear regression model, and when there are more than 1 predictor variables, it is called multiple linear regression

# Overview

- In statistics, we can describe how variables are related using a mathematical function. The function along with other assumptions is called a *model*. There are many models we can consider. In this class, we will focus on linear models, particularly, linear regression model.
- This lesson will also form the foundation for understanding more complex machine learning models.

# Linear Relationships

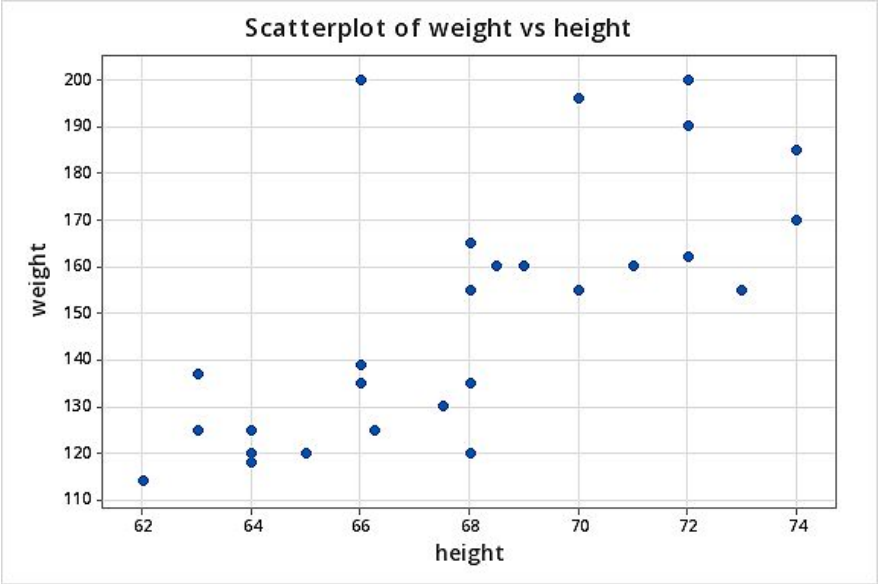
- To define a useful model, we must investigate the relationship between the response and the predictor variables.
- For a brief review of linear functions, recall that the equation of a line has the following form: $y = mx + b$
  - where m is the slope and b is the y-intercept.
- If the slope is positive, then there is a positive linear relationship, i.e., as one increases, the other increases. If the slope is negative, then there is a negative linear relationship, i.e., as one increases the other variable decreases. If the slope is 0, then as one increases, the other remains constant.

# Scatterplots

- If the interest is to investigate the relationship between two quantitative variables, one valuable tool is the scatterplot.
- A graphical representation of two quantitative variables where the explanatory variable is on the x-axis and the response variable is on the y-axis.

# Scatterplots

- The scatterplot shows that, in general, as height increases, weight increases. We say “in general” because it is not always the case. For example, the observation with a height of 66 inches and a weight of 200 pounds does not seem to follow the trend of the data.
- The two variables seem to have a positive relationship. As the height increases, weight tends to increase as well. The relationship does not seem to be perfectly linear, i.e., the points do not fall on a straight line, but it does seem to follow a straight line moderately, with some variability.



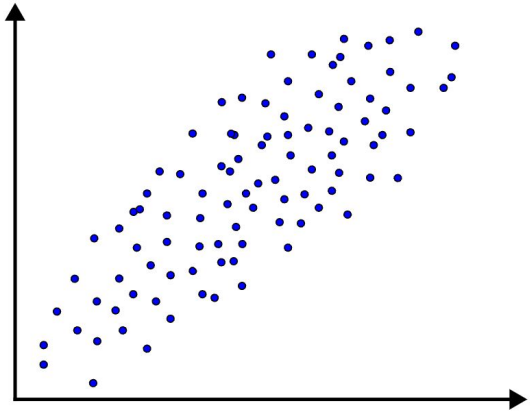
# Correlation

- If we want to provide a measure of the strength of the linear relationship between two quantitative variables, a good way is to report the correlation coefficient between them.
- The sample correlation coefficient is typically denoted as ***r***. It is also known as Pearson's ***r***. The population correlation coefficient is generally denoted as $\rho$, pronounced "rho."

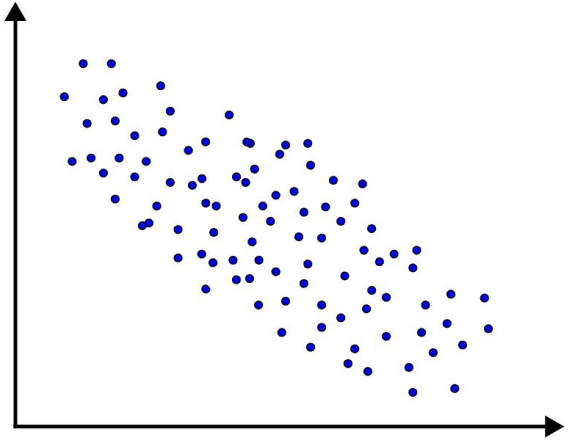
# Correlation Properties

- **$-1 \leq r \leq 1$**, takes values between -1 and +1, inclusive.
- The sign of the correlation provides the direction of the linear relationship. The sign indicates whether the two variables are positively or negatively related.
- A correlation of 0 means there is no linear relationship.
- There are no units attached to *r*.
- As the magnitude of r approaches 1, the stronger the linear relationship. As the magnitude of *r* approaches 0, the weaker the linear relationship.

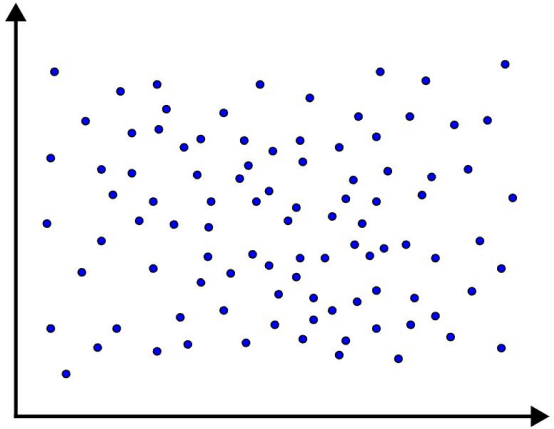
# Correlation Properties



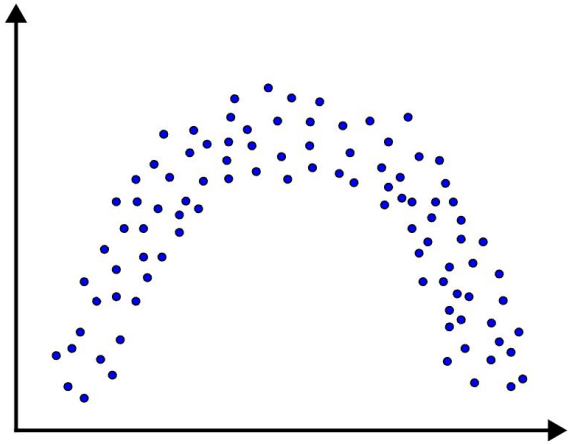
$r>0$



$r<0$



$r\approx0$



$r\approx0$

# Linear Regression

- The variable *Y* is the response variable and $y_i$ are the observed values of the response, *Y*.
- The variable *X* is the predictor variable and $x_i$ are observed values of the predictor, *X*.

The observations are considered as coordinates,$(x_i, y_i)$ for i= 1,....,n. As we saw before, the points,$(x_1,y_1)$.....$(x_n,y_n)$, may not fall exactly on a line, (like the weight and height example). There is some error we must consider.

**We combine the linear relationship along with the error in the simple linear regression model.**

# Linear Regression Model

The general form of the simple linear regression model is…

$$Y = \beta_0 + \beta_1 X + \epsilon$$

For an individual observation,

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

where,

- $\beta_0$ is the population y-intercept,
- $\beta_1$ is the population slope, and
- $\epsilon_i$ is the error or deviation of $y_i$ from the line, $\beta_0 + \beta_1 x_i$
- $y_i$ is the observed value of Y for the *i-th* data point
- $x_i$ is the corresponding X value

# Linear Regression Model

When we have multiple predictors, the linear regression model is denoted as:

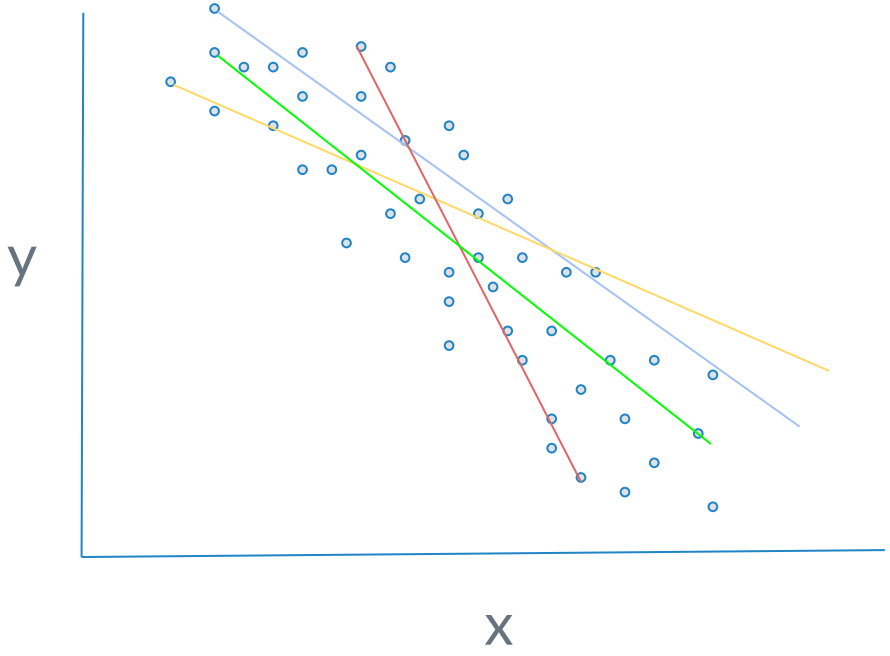
$$Y=\beta_0+\beta_1X_1+\beta_2X_2+\beta_3X_3+\cdots+\epsilon$$

Where:

- Y is the outcome (what we're trying to predict)
- $\beta_0$ is intercept (value of Y when all X's are 0)
- $\beta_1$, $\beta_2$, $\beta_3$, ... are slopes (effect of each variable on Y)
- $X_1$, $X_2$, $X_3$, ... are predictor (independent) variables
- $\epsilon$ is error (the random difference between actual and predicted Y)

# Linear Regression

Linear regression is a way to find the best line that summarizes the relationship between two things (variables).



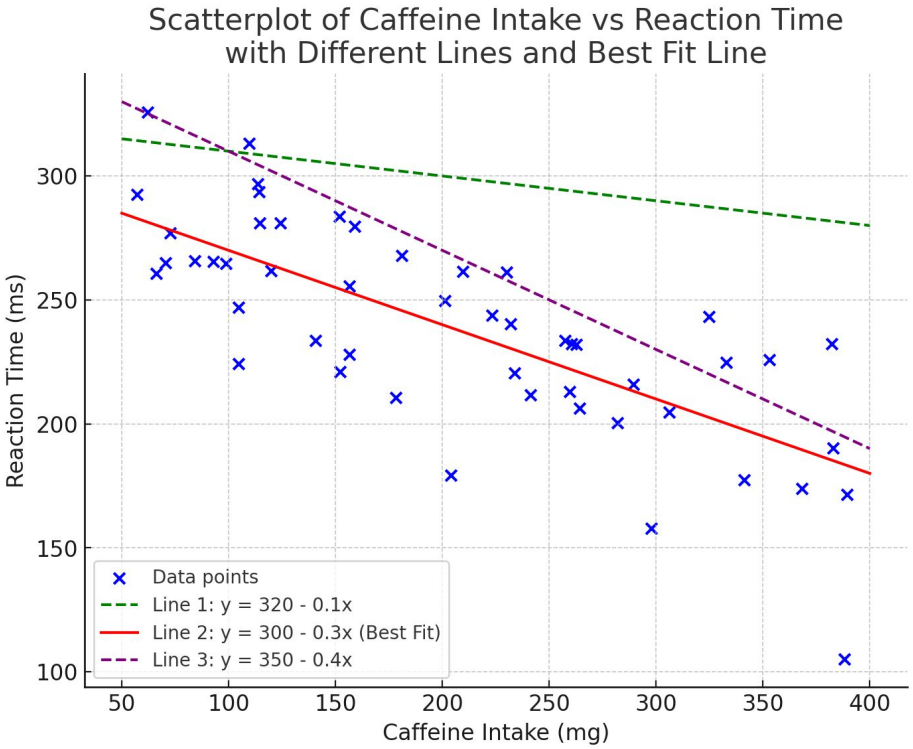
# Linear Regression

Example:

- We have data on caffeine and reaction time
  - predictor (independent): caffeine
  - response (dependent): reaction time
- 100 mg of caffeine (x) actually decreases reaction time (y) by 260 milliseconds, however, the predicted decrease in reaction time ($\hat{y}$) is 265 milliseconds.

# Linear Regression

The difference between the actual response time and predicted response time (265 - 260 = 5) is called the prediction error.



# Linear Regression

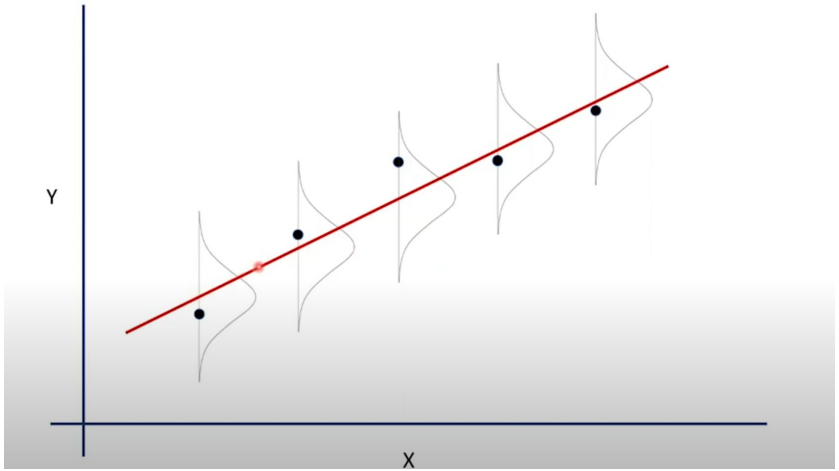
The "Least Squares":

To make sure we have the best line, we calculate the error for each observation and try to make the total error as small as possible. Instead of just adding up the errors, we square them to avoid negative and positive errors canceling each other out.

- For each observation, the squared prediction error is calculated.
- We add up all these squared errors, and the line with the smallest total squared error is considered the "best" line.

# Conditions of SLR

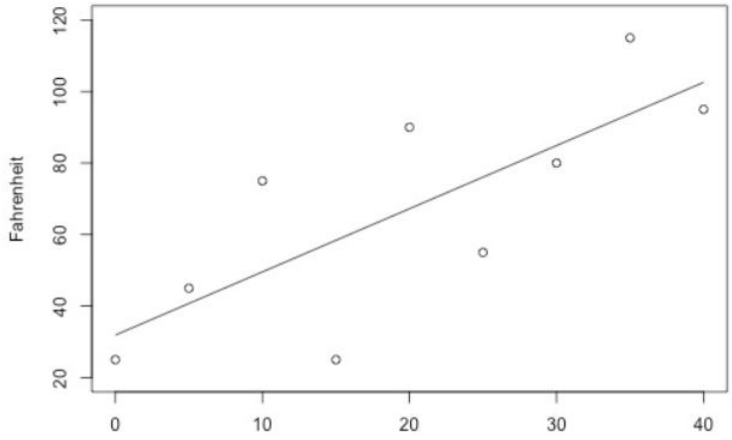
1. Linear Function: The mean of the response, E($Yi$) at each value of the predictor, $xi$, is a Linear function of the $xi$.
2. Independent: The errors, $\epsilon i$, are Independent.
3. Normally Distributed: The errors, $\epsilon i$, at each value of the predictor, $xi$, are Normally distributed.
4. Equal variances (denoted $\sigma 2$): The errors, $\epsilon i$, at each value of the predictor, $xi$, have Equal variances (denoted $\sigma 2$).



# Common Error Variance

- The common variance ($\sigma^2$) quantifies how much the data points deviate from the mean regression line.
- Variance ($\sigma^2$) helps us understand how precise our future predictions will be."
- Smaller variance indicates more precise predictions, as the data points stay closer to the estimated regression line.

A

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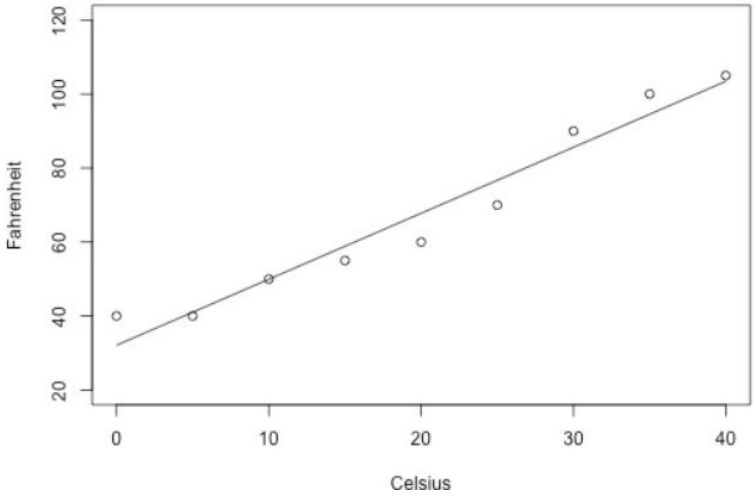


# Understanding R-Squared (Coefficient of Determination)

A statistical measure that explains how well one variable (predictor) explains another (response). Ranges from 0 to 1:

- 0: No explanation (weak relationship)
- 1: Perfect explanation (strong relationship)

Meaning of R-Squared Values

- R-squared = 0.68:
- "68% of the variation in the response variable is explained by the predictor."
- Good prediction, but still some randomness.

Important Note: Correlation vs. Causation

- High R-squared does not imply one variable causes the other.
- Example: Higher temperatures increase ice cream sales, but heat doesn't cause the purchase.

R-Squared in Different Fields

- Social Sciences: R-squared ~ 0.25 (complex human behavior)
- Engineering: R-squared ~ 0.80+ (more predictable systems)

# Pearson's Correlation Coefficient and R-squared

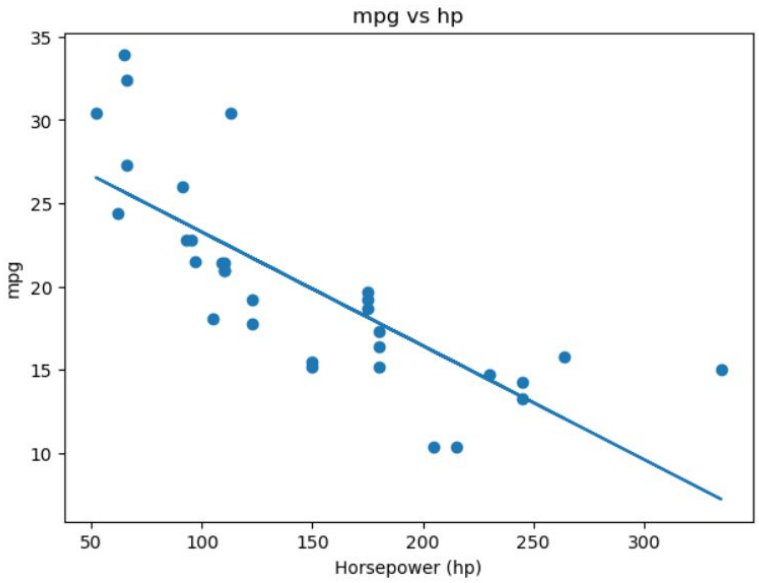
- The Pearson correlation coefficient (r ) provides information about the strength and direction of a linear relationship between two variables.
- R-squared quantifies how much of the variation in one variable can be explained by another and is derived from the Pearson correlation coefficient in simple linear regression.

$$R^2 = r^2$$

# Example - Simple Linear Regression

- Dataset: **mtcars**
  - The mtcars dataset contains fuel consumption and 10 aspects of car design/performance for 32 cars, including mpg, cyl, disp, hp, and more.
- **Question:**
  - Does the weight of a car significantly predict its fuel efficiency (measured in miles per gallon)?
- **Hypothesis:**
  - **Null Hypothesis ($H_0$):** There is no statistically significant relationship between car weight and fuel efficiency. $H_0$: $\beta_1=0$
  - **Alternative Hypothesis ($H_1$):** There is a statistically significant relationship between car weight and fuel efficiency. $H_1$:$\beta_1 \neq 0$

# Example - Simple Linear Regression




```
                            OLS Regression Results
==============================================================================
Dep. Variable:                    mpg   R-squared:                       0.602
Model:                            OLS   Adj. R-squared:                  0.589
Method:                 Least Squares   F-statistic:                     45.46
Date:                Tue, 21 Oct 2025   Prob (F-statistic):           1.79e-07
Time:                        17:21:21   Log-Likelihood:                -87.619
No. Observations:                  32   AIC:                             179.2
Df Residuals:                      30   BIC:                             182.2
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept     30.0989      1.634     18.421      0.000      26.762      33.436
hp            -0.0682      0.010     -6.742      0.000      -0.089      -0.048
==============================================================================
Omnibus:                        3.692   Durbin-Watson:                   1.134
Prob(Omnibus):                  0.158   Jarque-Bera (JB):                2.984
Skew:                           0.747   Prob(JB):                        0.225
Kurtosis:                       2.935   Cond. No.                         386.
==============================================================================
```